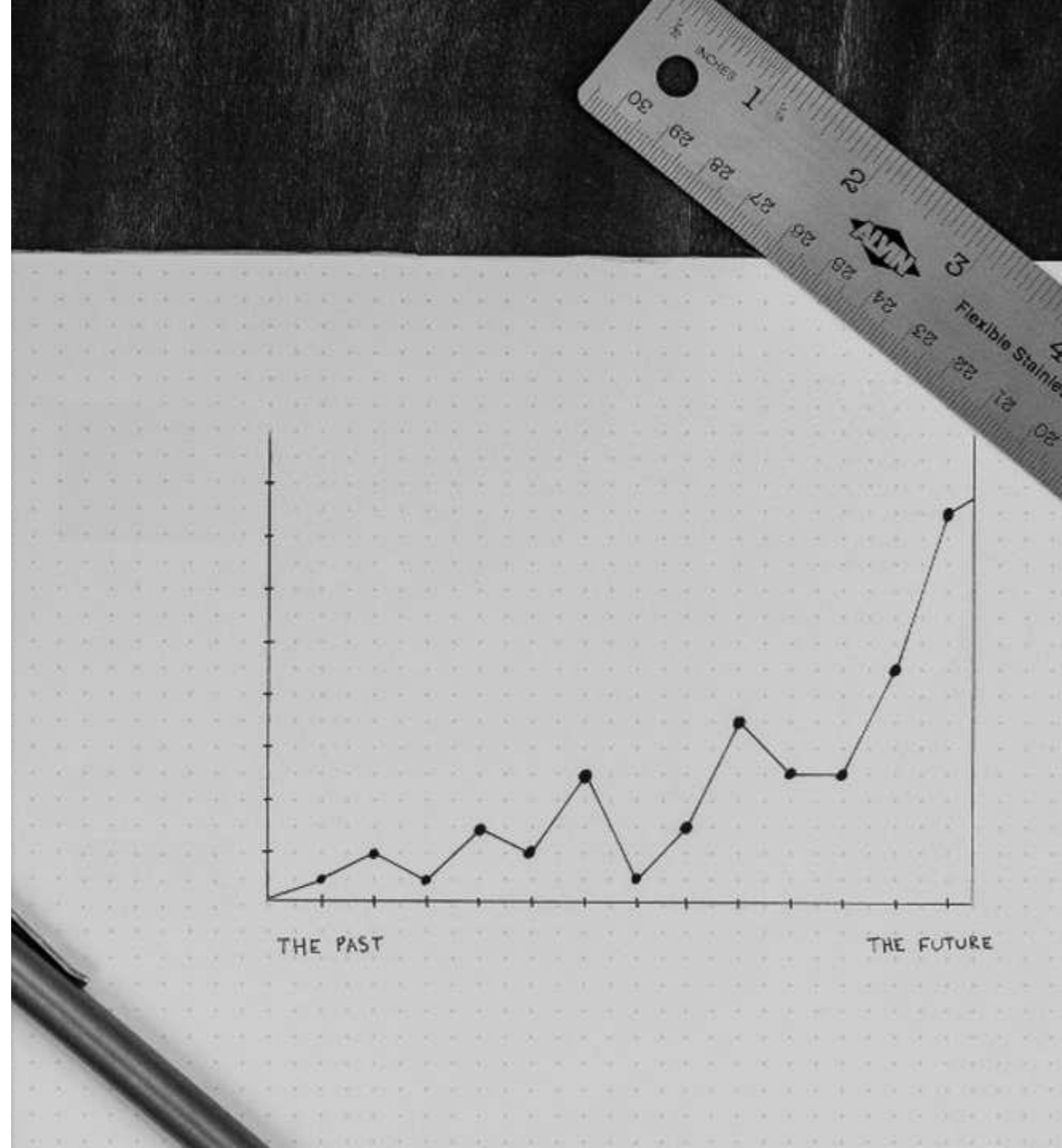




Designing dashboard

Dashboard builder



DashBoard Builder

In Semarchy xDM Dashboards, designers can:

- Connect to a variety of datasources, including **data warehouses, enterprise applications** or **Semarchy xDM** data locations.
- Develop **queries** on those data sources. These queries return datasets with identified attributes.
- **Design charts** that use the query attributes as dimensions or aggregated metrics to render visualizations, such as **bar, column, line** or **pie charts**.
- Arrange charts into **dashboards**, which are grouped into **dashboard applications**.
- Optionally **embed charts** and dashboards into Semarchy xDM forms and applications.

Dashboard Builder

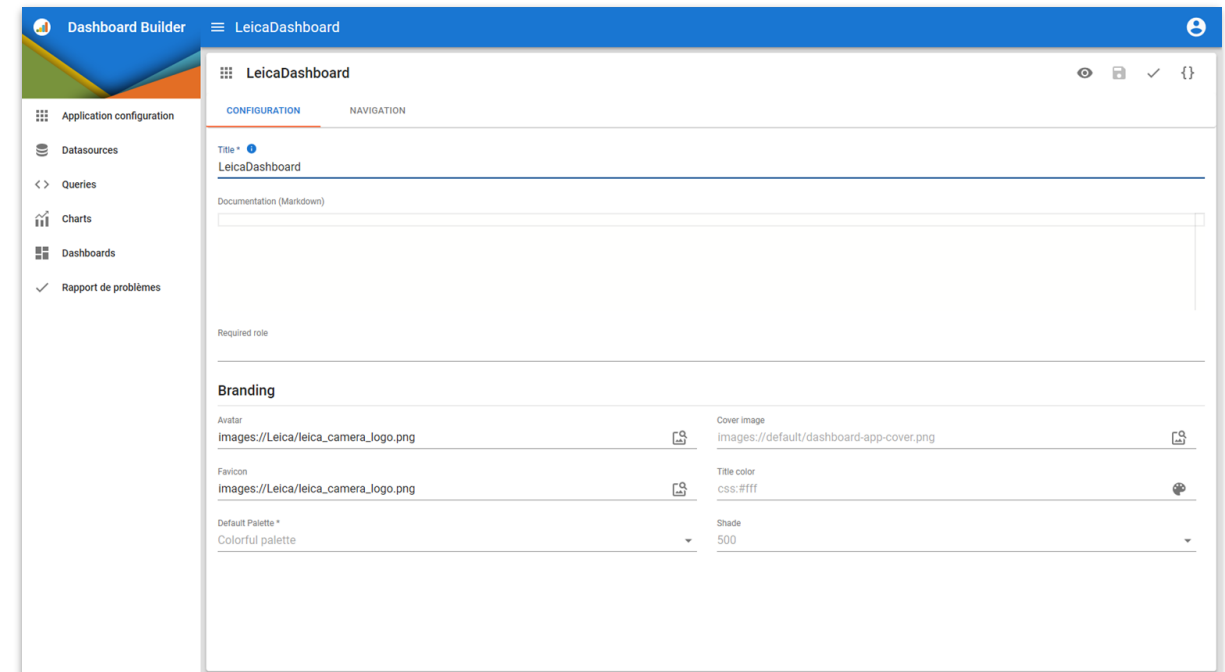
To create an application:

1. In the application list, click on the **Create application** button.



1. Enter a name for your application and then click **Create**

1. In this editor, configure the application properties:



Connecting your data

- Semarchy xDM Dashboards connect to your data via **datasources**.
- Datasources must be defined in Dashboard Builder prior to creating **queries** on top of them.
- A **Platform Datasource** is declared from the **Configuration** section

New Datasource

Platform Datasource*

SEMARCHY_CUSTOMER_B2C_MDM ▼

Name

Customer360

CANCEL CREATE

Query

To create a query:

1. Select **Queries** in the navigation drawer.
2. Click on the Create new query floating action button.
3. In the query editor, enter the SQL
4. Use the Run Query button in the toolbar.
5. Configure each attribute in the Query Attributes table

Define whether this query attribute may be used as a Measure or Dimension in your charts.

Dashboard Builder CMAVesselsDashboards > Queries > AttributeValuesComparisonCount

AttributeValuesComparisonCount

Datasource* CmaVesselMdm

SQL Query *

```
select 'OPERATOR' as attribute, count(a.IMO_CODE) as differences_count from CMA_VESSEL_MDM.md_vessel a, CMA_VESSEL_MDM.md_vessel b
where a.IMO_CODE = b.IMO_CODE
and a.B_PUBID = 'VCH'
and b.B_PUBID = 'CDM'
and a.OPERATOR <> b.OPERATOR
UNION
```

Press CTRL+Enter or ⌘+Enter to run the query.

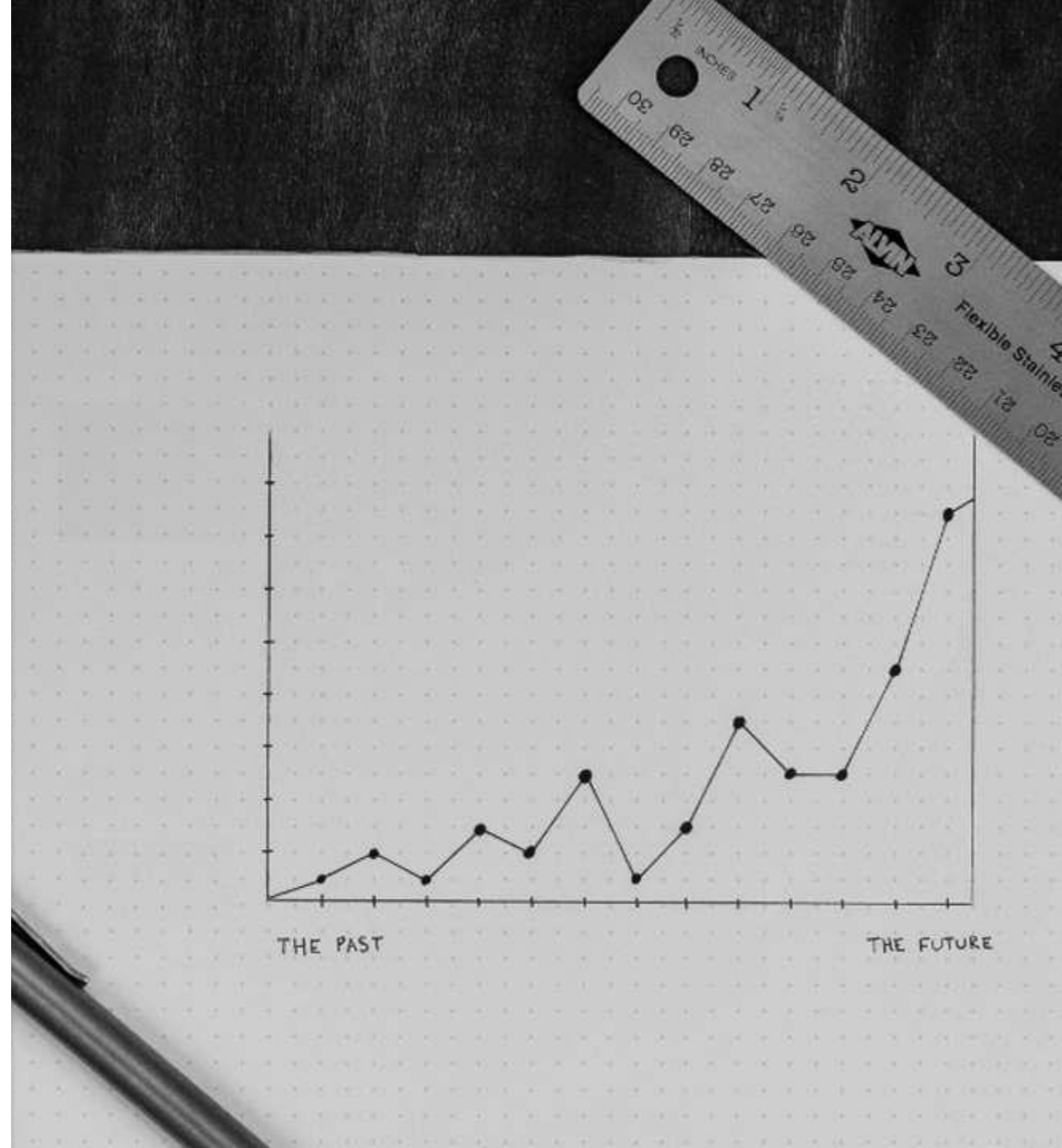
Query attributes

Name	Label	Datatype	Format	Measure	Dimension	Time dimension
ATTRIBUTE		Text		☒	☒	☒
DIFFERENCES_COUNT		Number	1 000 000,00	☒	☒	☒



Designing dashboard

Dashboard application



Building chart

- **Measures** and **Dimensions**, defining a **Pivot Query**, that is the shape of the dataset rendered by the chart. Measures and dimensions are based on query attributes.
- A **Chart Type**, defining the visualization used to display the dataset. For example, a line chart.
- **Configuration** elements applies to the measures, the dimensions and to the chart itself. These elements depend on the chart type.
- The **measures** and **dimensions** of a chart define the pivot query. A good representation of a pivot query is in the form of a table:
- The **rows** (aka. categories) and **columns** (aka. series) of the table are the dimensions, the measures are the values in the cells at the intersection of the rows and columns.
- All objects in Dashboard Builder are stored in a **YAML** form so you configure or share them.

Chart type - Simple charts

- **Number chart** displays a single number value
- **Pivot Table** displays the pivot query as a table with a hierarchy of rows and columns.
- **Donut** or **Pie chart** show percentages of values between categories, by dividing a circle or a disk into proportional segments.
- **Gauge chart** shows values as performance gauges for each category, with lower and upper ranges.

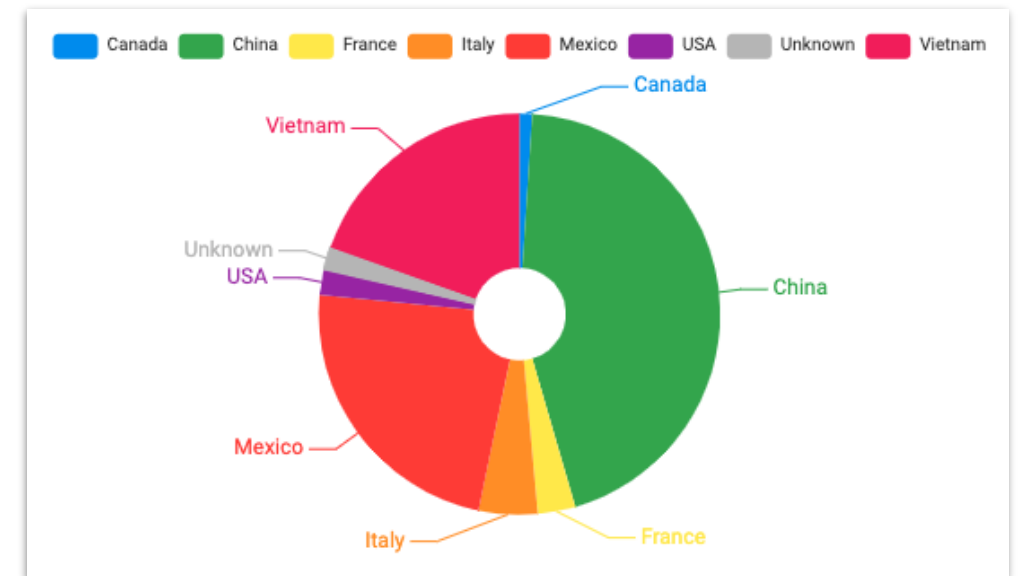
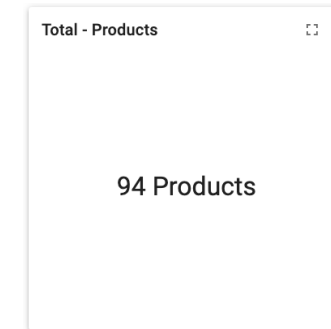


Chart type - Axis charts

Axis Charts are used to compare values across many categories:

- Column and Bar charts show values side by side as vertical or horizontal bars, across categories. These charts support clustering, supporting two or more data series plotted side-by-side and grouped together under categories.
- Line graph shows values over a continuous interval. This chart is most frequently used to show trends and data changes over time. It supports multiple data series to compare values and trends.
- Area graph is similar to the line graph with the area below the line filled. Area graphs support stacking.
- Combined chart use a secondary value axis to combine bar/columns and line charts on the same visualization.

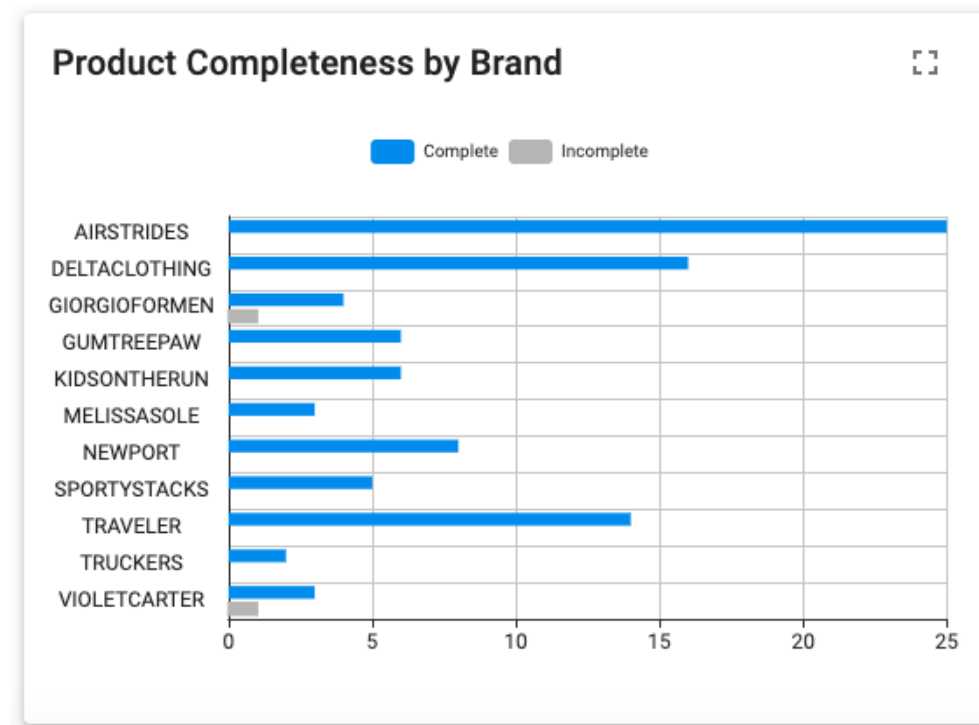
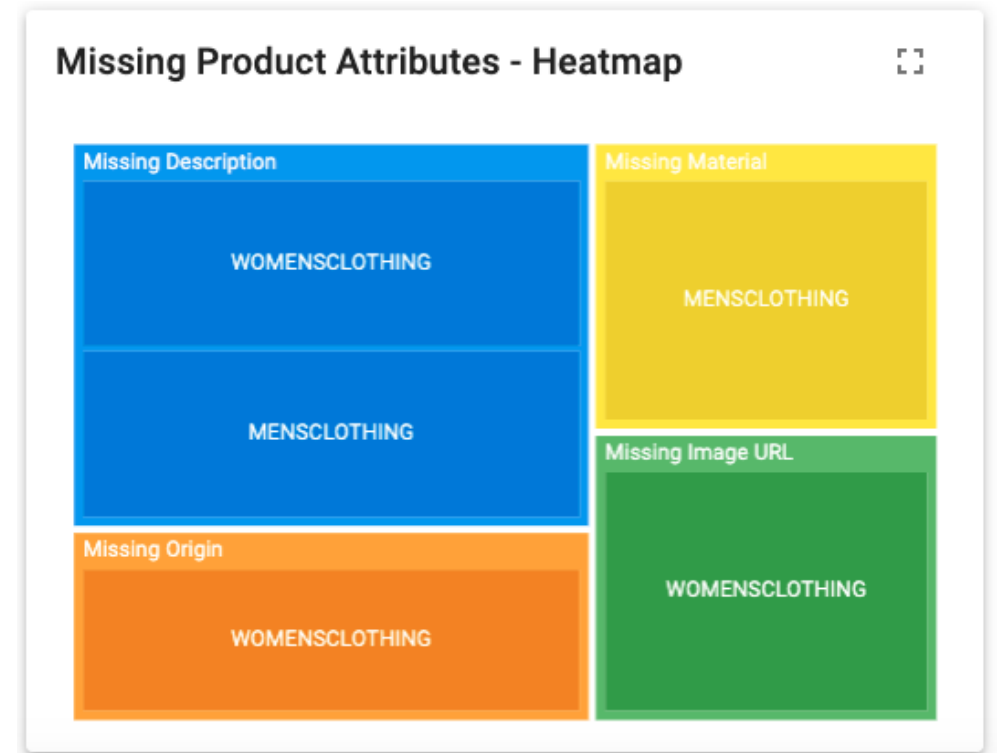


Chart type - Hierarchical charts

- Sunburst diagram shows a hierarchical distribution of the values in a series of rings.
- Treemap shows a hierarchy of categories while also displaying quantities for each category via area size.



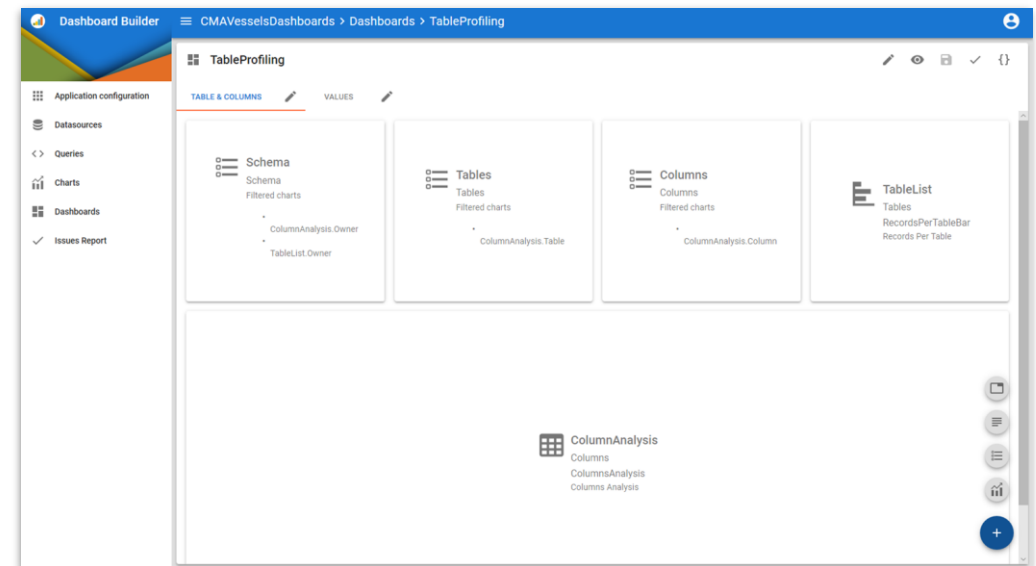
Dashboard

A **dashboard** is composed of one or multiple **tabs**, each tab is composed of elements of the following types:

- **Chart** elements, pointing to existing charts.
- **Text** elements, containing rich text content.
- **Slicer** elements used to filter **charts** content in the **dashboard**.

Slicers are used to filter the content of the **charts** in the **dashboard**.

Each **slicer** is linked to one or multiple **charts** via a chart parameter.



Embedded chart

Chart can be embedded in form (with parameters)

Add form chart

Properties Progress Validation Report

Form Chart: Chart

Name: Chart

Use Custom Label: ☐

Label:

Description:

Documentation:

Dashboard Application: IntelligentDataHub

Chart: UnitPriceEvolution

Chart definition

CustomerForm Form (Customer)

Attributes: n+l

Name	Label	Value	Configuration
FormTab : Main	Main		
Image		Image	
CustomerID		CustomerID	
CustomerName		CustomerName	
TotalRevenue		TotalRevenue	
CustomerNamePhonetic		CustomerNamePhonetic	
InputAddress		InputAddress	
GeoCodeAddress		GeoCodeAddress	
IsValid		IsValid	
Chart			

Chart parameter

Properties Progress Validation Report

Form Chart: Chart

Name and Definition

Display Properties

Chart Parameters

System Properties

Name	Value Interpreter	Value
Customer_ID	SemQL	CustomerID

Chart component

Embedded dashboard

- Dashboard can be **embedded** in form and in the **xDM application**.
- Possibly **binding** entity attribute values to chart and **dashboard parameters**.
- Dashboards can be open as **embedded editors**, using a dedicated application action.

